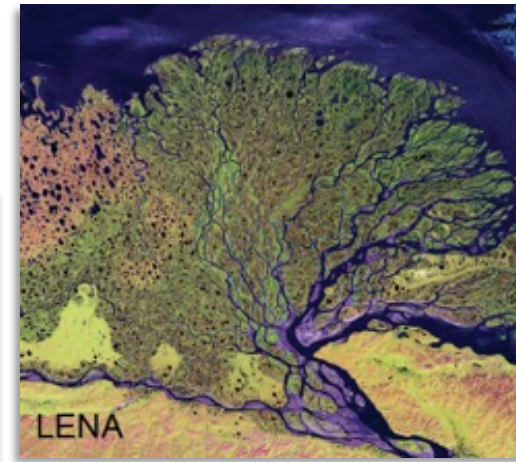
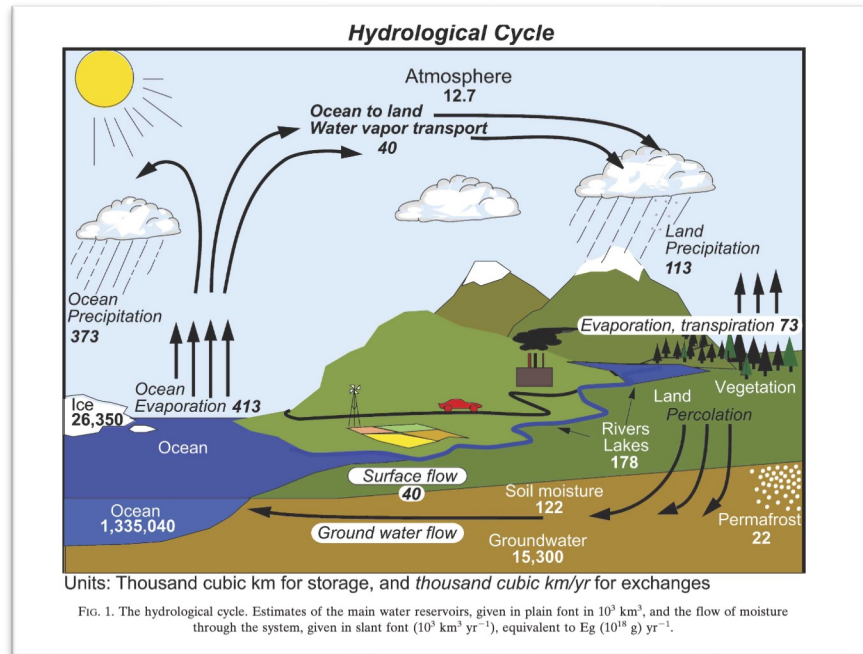


Topic #3: Land Water Storage Variability

- **Topic: Land water storage variability as a function of human and natural controls**
- Datasets: GRACE water storage (equivalent water height over land), TRMM precipitation, GPM precipitation, AIRS near- surface temperature, ECMWF ERA5 near-surface temperature, GLDAS soil moisture (layer1), SMAP soil moisture
- Geographic foci: Global, Northern India, southwest US
- Questions:
 - Is there an apparent seasonality in GRACE water storage or GLDAS/SMAP soil moisture in heavily managed agricultural regions? How does that seasonality temporally align with seasonality in TRMM/GPM precipitation and AIRS near-surface temperature, and align with the agricultural seasons when soil water is used for irrigation?
 - Is there longer-term variability in GRACE time series that are not represented in TRMM/GPM precipitation, AIRS near-surface temperature or GLDAS/SMAP soil moisture (e.g. interannual & trend)?
 - From these datasets, what conclusions can we draw regarding the natural or anthropogenic drivers of water storage and soil moisture in this system at different time scales? Can these datasets be used to identify and quantify human influences on the water cycle?
- **Contact Scientist: JT Reager (John.Reager@jpl.nasa.gov)**

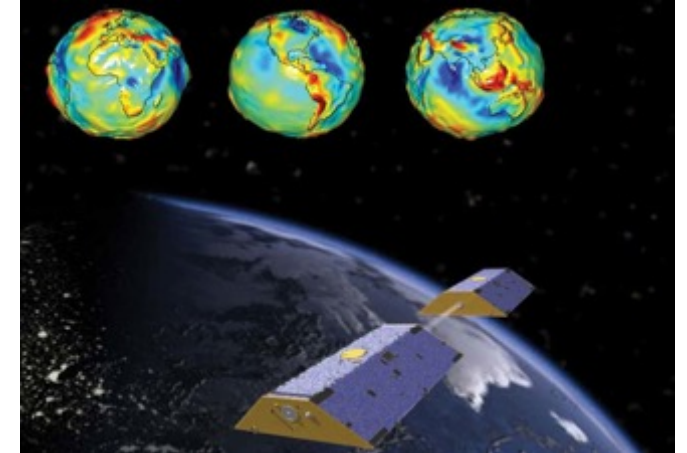
Hydrology is hard

- Not many observations
- Highly variable in space and time
- Observational uncertainty affects model accuracy
- Our understanding of the global water cycle portrays a stable and unchanging system
- Remote sensing is the perfect tool



Gravity Recovery and Climate Experiment: GRACE and GRACE-FO

- First mission to see hydrology beneath the surface!
- Launched 2002: long record is critical for good science
- Still flying and returning data, with periodic outages to conserve power
- Measures total integrated water storage change beneath the satellites
- ~2-3 degree resolution
- GRACE-FO launch 2017



nature
climate change

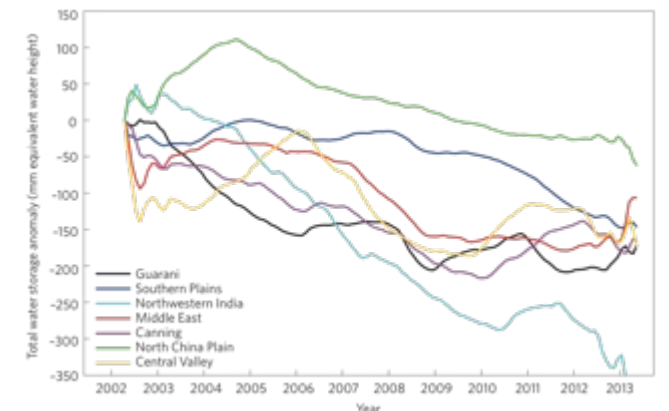
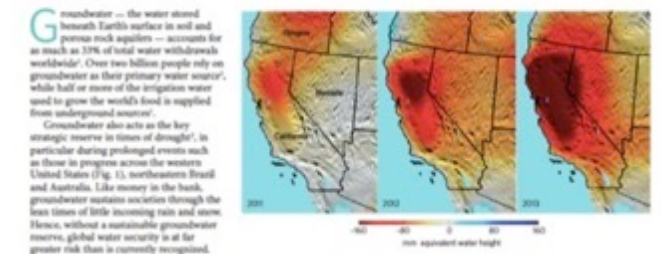
29 October 2014

opinion & comment

COMMENTARY: The global groundwater crisis

J. S. Famiglietti

Groundwater depletion the world over poses a far greater threat to global water security than is currently acknowledged.



Groundwater...

...accounts for 96% of unfrozen freshwater globally

...is the major source of water for over 2 billion people globally

...supplies roughly 45% of water for irrigation globally

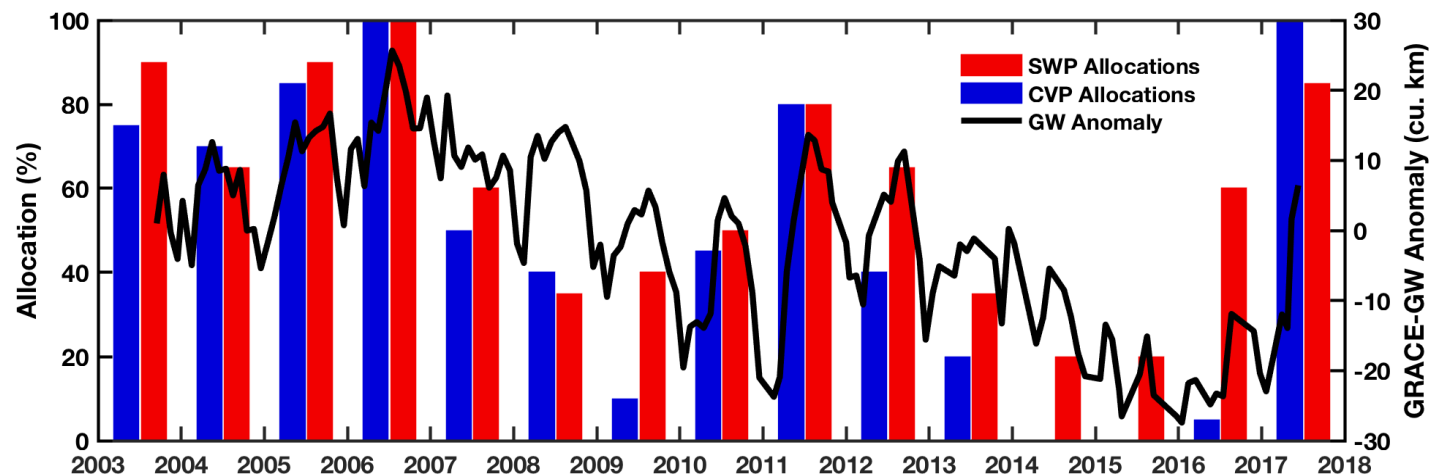
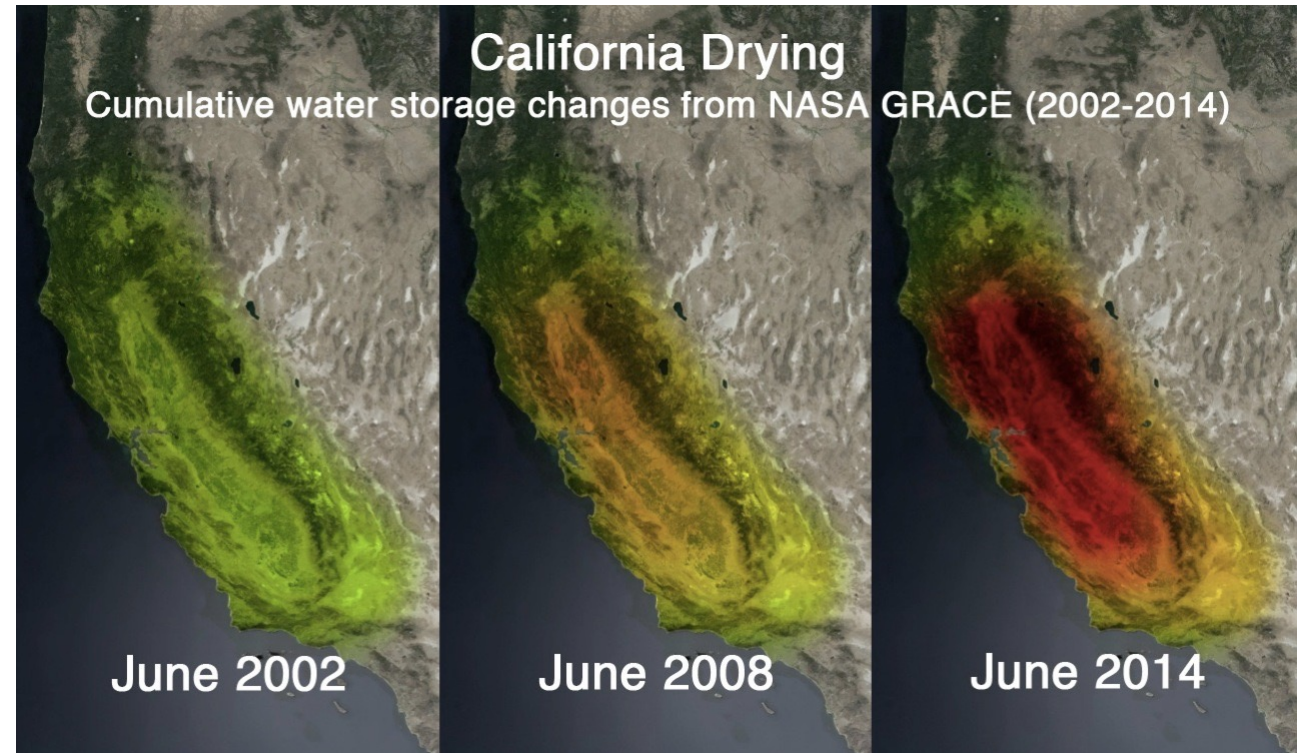
...supplies over half of the drinking water in the United States

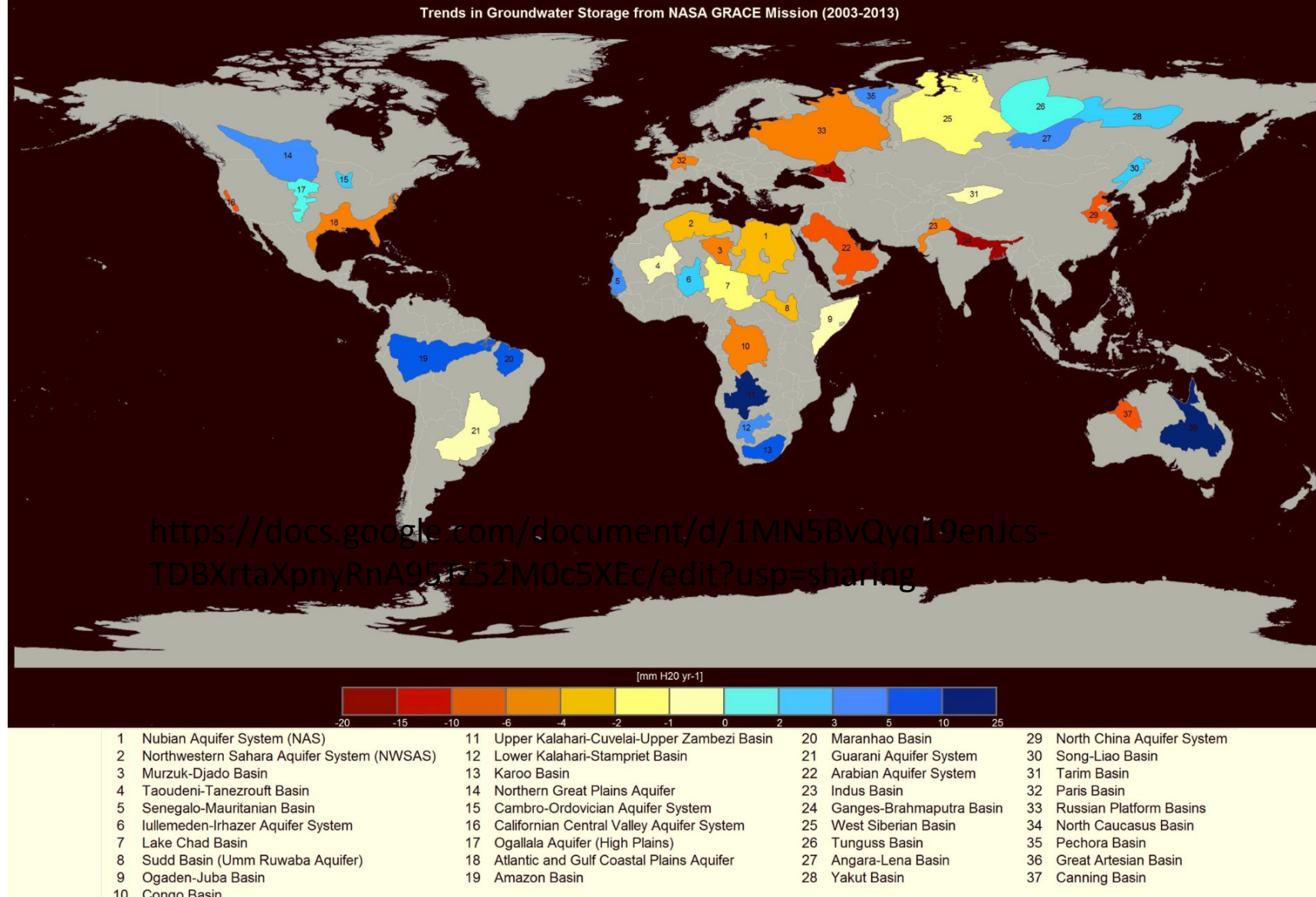
...supplies over 60% of the water used for irrigation in the United States

...is THE strategic water reserve in times of extreme drought

However, groundwater withdrawals go unmonitored and unmanaged in most places around the world, including most states in the United States, and the lack of management is becoming apparent. This poses a considerable threat to our nation's water security







Richey et al., 2014

37 largest aquifers

21 are declining

13 are being rapidly depleted

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